

Homework 1. Introduction to Forecasting

You can get up to 10 points for this homework.

Question 1 (3 points). You are the administrator of a large metropolitan hospital that operates the only 24-hour emergency room in the area. You must schedule attending physicians, resident physicians, nurses, laboratory, and support personnel to operate this facility effectively.

- ☐ A What measures of effectiveness do you think patients use to evaluate the services that you provide?
- ☐ B How are forecasts useful to you in planning services that will maximise these measures of effectiveness?
- ☐ C What planning horizon do you need to use? Does this lead to short-range or long-range forecasts?

Question 2 (3 points). Consider an airline that operates a network of flights that serves 200 cities in the continental United States.

- ☐ A What long-range forecasts do the operators of the airline need to be successful?
- ☐ B What forecasting problems does this business face on a daily basis?
- ☐ C What are the consequences of forecast errors for the airline?

Question 3 (4 points). Your company manufactures and distributes soft drink beverages, sold in bottles and cans at retail outlets such as grocery stores, restaurants and other eating/drinking establishments, and vending machines in offices, schools, stores, and other outlets. Your product line includes about 25 different products, and many of these are produced in different package sizes.

- ☐ A What forecasts do you need in this business to be successful?
- ☐ B Is the demand for your product likely to be seasonal? Explain why or why not?
- ☐ C Does the shelf life of your product impact the forecasting problem?
- ☐ D What data do you think that you would need to be able to produce successful forecasts?